FIGHTING INFECTIONS

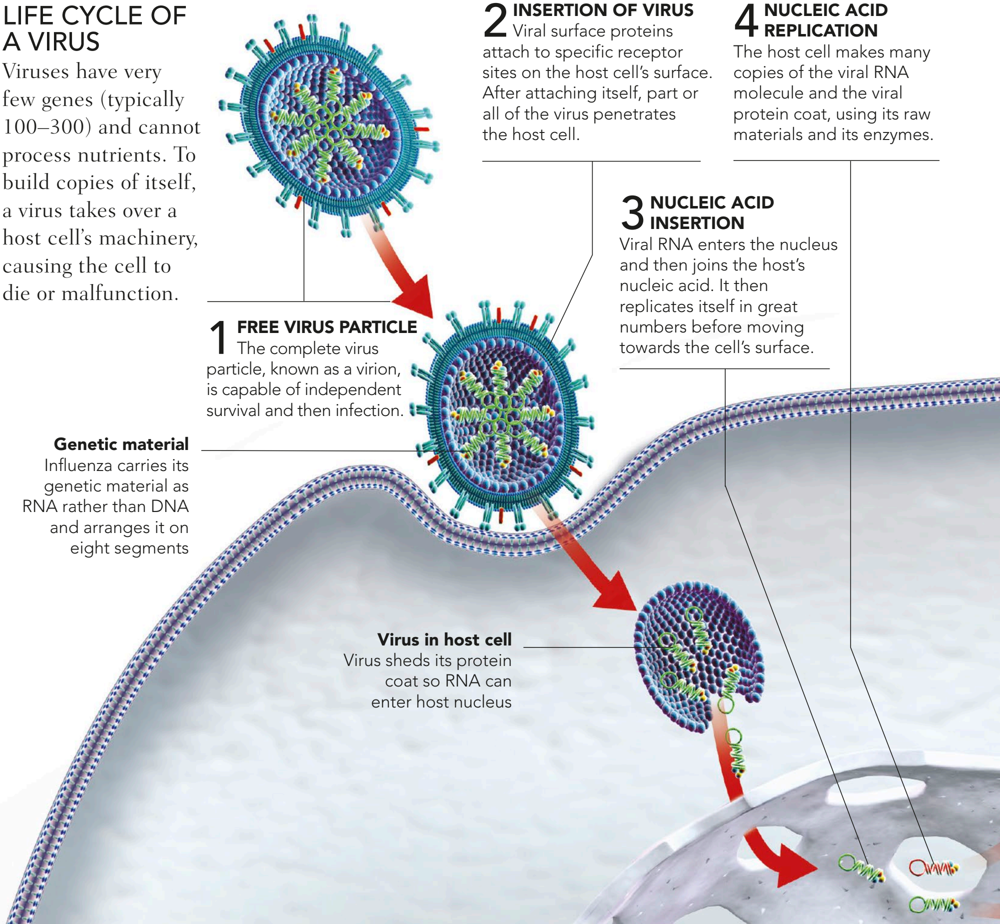
AN INFECTION OCCURS WHEN MICROORGANISMS ENTER THE BODY, THEN SURVIVE, MULTIPLY, AND DISRUPT CELL FUNCTION. THE INFECTION MAY BE LOCAL, SUCH AS IN A WOUND, OR SYSTEMIC, IN WHICH MANY PARTS OF THE BODY ARE AFFECTED.

VIRUSES

Viruses are the smallest microbes; millions would cover the head of a pin. Many types of virus can stay inactive for long periods and survive freezing, boiling, and chemical attack. Yet they can activate suddenly when an opportunity of invading a living cell arises. Viruses are obligate parasites, which means they must have living cells, or host cells, in order to replicate themselves. The typical virus particle has a single or double strand of genetic material (nucleic acid – either DNA or RNA) surrounded by a shell-like coat of protein (capsid), and sometimes a protective outer envelope.

LIFE CYCLE OF A VIRUS

Viruses have very few genes (typically 100–300) and cannot process nutrients. To build copies of itself, a virus takes over a host cell's machinery, causing the cell to die or malfunction.



1 FREE VIRUS PARTICLE The complete virus particle, known as a virion, is capable of independent survival and then infection.

2 INSERTION OF VIRUS Viral surface proteins attach to specific receptor sites on the host cell's surface. After attaching itself, part or all of the virus penetrates the host cell.

3 NUCLEIC ACID INSERTION Viral RNA enters the nucleus and then joins the host's nucleic acid. It then replicates itself in great numbers before moving towards the cell's surface.

4 NUCLEIC ACID REPLICATION The host cell makes many copies of the viral RNA molecule and the viral protein coat, using its raw materials and its enzymes.

Genetic material Influenza carries its genetic material as RNA rather than DNA and arranges it on eight segments

Virus in host cell Virus sheds its protein coat so RNA can enter host nucleus